Parameter Estimation Criteria

Copyright © Barry Van Veen 2014

AllSignalProcessing.com

Key Concepts

1) Parameter estimation assumes the observed data depends on an unknown parameter of interest and that this dependence is captured by a probability density function.

2) The estimator of the unknown parameter of interest is a function of the data.

3) Two common criteria for finding the function that estimates the parameter are the maximum likelihood and Bayesian criteria.

4) Maximum likelihood estimation seeks to find the parameter that maximizes the probability density function evaluated at the observed data.

    a) Assumes the parameter is deterministic but unknown.

    b) Results in an optimization problem that can be solved using analytic or numerical techniques.

    c) Is asymptotically (as number of data measurements approaches ∞) unbiased.

    d) Achieves the smallest possible variance of any unbiased estimator.

5) Bayesian estimation involves the posterior probability density function of the unknown parameter given the observed data.

    a) Assumes the parameter is random with a known prior probability density function.

    b) Powerful framework for incorporating prior information about the parameter using the prior probability density function.

    c) Minimizes the expected value of a non negative cost function of the error between the estimate and true unknown parameter.

    d) Mean-squared error cost function leads to the mean of the posterior probability density.

e) Absolute error cost function leads to the median of the posterior probability density.

f) Uniform error leads to the maximum of the posterior probability density (maximum a posteriori or MAP estimator).

Video Screenshots

0:20

Parameter estimation: observed data $\underline{x}$ depends on 1 unknown parameter(s) $\underline{\alpha}$ through pdf

$$\underline{x} \sim f(\underline{x}; \underline{\alpha})$$

We seek estimator $\hat{\alpha} = g(\underline{x})$ – how can we find $g(\cdot)$?

Two common criteria

1) Maximum likelihood given observation $\underline{x}_0$,

$$\hat{\underline{\alpha}} = \arg\max_{\underline{\alpha}} \{f(\underline{x}_0; \underline{\alpha})\}$$

Assumes $\underline{\alpha}$ deterministic but unknown

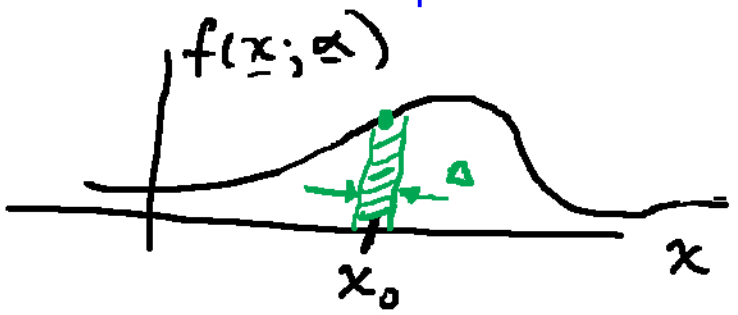
2) Bayesian use posterior pdf $f(\underline{\alpha} | \underline{x}_0)$

Assumes $\underline{\alpha}$ random, requires prior pdf $f(\underline{\alpha})$

Maximum Likelihood.

2:10

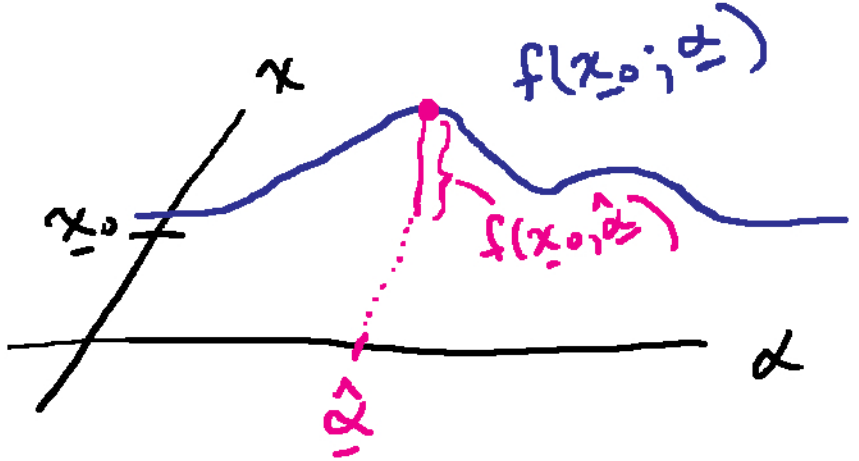
Probability of observing data $\underline{x}$ near $\underline{x}_0$



$$\int_{x_0-\Delta/2}^{x_0+\Delta/2} f(\underline{x};\underline{\alpha})\,d\underline{x} \approx f(\underline{x}_0;\underline{\alpha})\,\Delta$$

for Δ small

MLE finds the $\underline{\alpha}$ that gives max probability of observing $\underline{x}_0$



$$\max_{\underline{\alpha}} f(\underline{x}_0;\underline{\alpha})$$

Why MLE?

4:29

1) Straightforward approach - analytic or numerical optimization problem

2) Asymptotic properties (number of measurements $\to \infty$)

   a) MLE is unbiased $E\{\hat{\underline{\alpha}}_{MLE}\} = \underline{\alpha}$

   b) MLE achieves smallest possible variance of any unbiased estimator

6:24

Bayesian Estimation

Given model $f(\underline{x}|\underline{\alpha})$ and prior $f(\underline{\alpha})$, find posterior pdf $f(\underline{\alpha}|\underline{x})$

Bayes' rule: $f(\underline{\alpha}|\underline{x}) = f(\underline{x}|\underline{\alpha}) f(\underline{\alpha}) / f(\underline{x})$

Estimator: $\hat{\underline{\alpha}} = \arg\min_{\hat{\underline{\alpha}}} E\{C(\underbrace{\hat{\underline{\alpha}} - \underline{\alpha}}_{\text{error}})\}$ ← nonnegative cost fcn

Specific cases:

1) mean squared error $C(\hat{\underline{\alpha}} - \underline{\alpha}) = |\hat{\underline{\alpha}} - \underline{\alpha}|^2 \Rightarrow \hat{\underline{\alpha}} = \text{mean}\{f(\underline{\alpha}|\underline{x})\}$

2) absolute error $C(\hat{\underline{\alpha}} - \underline{\alpha}) = |\hat{\underline{\alpha}} - \underline{\alpha}| \Rightarrow \hat{\underline{\alpha}} = \text{median}\{f(\underline{\alpha}|\underline{x})\}$

3) max a posteriori (MAP) $C(\hat{\underline{\alpha}} - \underline{\alpha}) = \begin{cases} 1, \hat{\underline{\alpha}} \neq \underline{\alpha} \\ 0, \hat{\underline{\alpha}} = \underline{\alpha} \end{cases} \Rightarrow \hat{\underline{\alpha}} = \arg\max_{\underline{\alpha}} \{f(\underline{\alpha}|\underline{x})\}$

Why Bayesian?

10:06

1) Possible to find true optimum estimates, like minimum MSE

2) Powerful framework for incorporating prior information (e.g. physical constraints)

Can be difficult to find $f(\underline{\alpha}|\underline{x})$.

For more great educational signal-processing content:

AllSignalProcessing.com

- Intuitive, easy-to-follow style
- Organization that saves you time
- Answers to your questions
- Learn at your pace
- Award winning instruction